YE TIAN

Ph.D. Candidate in Computer Science, UC San Diego
Multimodal AI · LLM Agents · 3D Perception · Efficient & Adaptive ML
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RESEARCH PROFILE

My research focuses on **multimodal AI, LLM agents, and efficient AI systems**, aiming to build intelligent systems that can perceive, reason, and adapt in real-world environments. My work integrates **multimodal perception with LLM-driven reasoning** to enable complex understanding and decision-making across heterogeneous data sources. I also investigate **online and continual adaptation** to improve model robustness under dynamic conditions, alongside **resource-efficient learning and inference** for deployment on edge devices. These efforts span applications in **autonomous systems, personalized intelligence, and edge AI**.

EDUCATION

UCSD, Department of Computer Science and Engineering — ***Sep. 2024 – Present***
Advisor: IEEE/ACM Fellow Prof. Tajana Rosing — Ph.D.

USTC, College of Computer Science and Technology — ***Sep. 2021 – Jun. 2024***
Advisor: IEEE/ACM Fellow Prof. Xiang-Yang Li — M.S.

Northwest A&F University, College of Information Engineering — ***Aug. 2017 – Jun. 2021***
Advisor: Prof. Yong Deng and Prof. Bingyi Kang — B.S.

TECHNICAL SKILLS

AI & Machine Learning: Deep Learning · Multimodal Learning · Computer Vision · 3D Perception · Sensor Fusion · Time-Series Modeling.

LLMs & Agentic AI: Large Language Models (LLMs) · LLM Agents · Retrieval-Augmented Generation (RAG) · Tool-Augmented Reasoning · Knowledge Graphs · Multimodal Reasoning.

Efficient & Adaptive AI: Efficient Learning and Inference · Online/Test-Time Adaptation · Continual Learning · On-Device AI · Federated Learning · Hyperdimensional Computing (HDC).

Programming & Systems: Python · PyTorch · TensorFlow · OpenCV · Linux · Git · Docker · Kubernetes · AWS · Azure.

INTERNSHIP EXPERIENCE

NAIST (Japan) – Research Intern, Interactive Media Design Laboratory. ***Jul. 2024 – Sep. 2024***
Advisor: Prof. Hirokazu Kato
Research on **AR/VR-based human–computer interaction**, integrating **multimodal perception** and mobile computing for intelligent interactive experiences.

IDATA – Research Intern, Intelligent Perception Research Center. ***Oct. 2023 – May 2024***
Advisor: Prof. Xiang-Yang Li
Research on **computer vision and intelligent document understanding** for automated recognition, structural analysis, and generation of industrial electrical diagrams.

Daqo Group Industry – Research Intern, AI Research Department. ***Oct. 2022 – Oct. 2023***
Advisor: Prof. Jiahui Hou
Developed **computer vision-based perception methods** for electrical diagram component detection, circuit recognition, and structured diagram understanding to support **industrial AI automation**.

DAII – Research Intern, Smart City Algorithm Design Group. ***Apr. 2021 – Sep. 2021***
Advisors: Prof. Yanyong Zhang and Prof. Hao Zhou
Research on **IoT device authentication and embedded sensing**, including secure device identification and deployment of reliable IoT interaction platforms.

SELECTED RESEARCH PROJECTS

Multimodal AI and LLM Agents for Personalized Intelligence *Sep. 2024 – Present*

- **DailyLLM: Multimodal Sensing and Context Understanding.** Developed an LLM-based framework that transforms heterogeneous smartphone and smartwatch sensor data into context-aware activity logs and natural-language summaries. Achieved **17%** higher precision than a 70B-parameter SOTA model using only a 1.5B-parameter LLM, with nearly **10**× faster inference and on-device deployment on Raspberry Pi 5 ***(SenSys'25, MASS'25)***.
- **LifeAgentBench & LifeAgent: Long-Horizon Reasoning with LLM Agents.** Developed a benchmark comprising 22,573 questions to evaluate long-horizon and cross-dimensional reasoning across lifestyle health data, systematically assessing 13 LLMs. Designed a **tool-augmented LLM agent** that integrates evidence retrieval, deterministic computation, and task decomposition to improve reasoning reliability on challenging queries ***(EMNLP'26)***.

Multimodal Reasoning and Adaptation for Autonomous Systems *Jun. 2025 – Present*

- **KROMA: Knowledge-Retaining Online Multimodal Adaptation.** Developed a scene-aware online adaptation framework for robust 3D object detection under changing driving conditions. Designed a **lightweight scene recognition module** and **context-indexed model memory** to retrieve and adapt historical model states, enabling adaptation to new environments while mitigating **catastrophic forgetting**. Achieved **19.31%** improvement in overall detection performance across diverse driving scenarios ***(CVPR'27, in submission)***.
- **KLDrive: Knowledge-Graph-Grounded 3D Scene Reasoning.** Developed a **multimodal 3D scene knowledge graph** from LiDAR and camera data through 3D object perception, spatial relationship modeling, and temporal association. Designed a **knowledge-graph-augmented LLM agent** integrating structured retrieval and multi-step reasoning for fine-grained autonomous driving scene understanding and decision-making ***(AAAI'27, under review)***.
- **DroneFL: Federated Learning for Multi-UAV Target Tracking.** Designed a **lightweight federated learning framework** for collaborative UAV tracking, using position-invariant trajectory prediction and altitude-aware normalization to address heterogeneous onboard observations. Reduced prediction errors by **6**–**83%** and tracking distance by **0.4**–**4.6%**, while supporting real-time inference on Raspberry Pi 5 ***(T-ASE'26, under review)***.

Efficient and Adaptive AI for Resource-Constrained Systems *Oct. 2024 – Sep. 2025*

- **FHDnn: Communication-Efficient Federated Learning.** Developed a **hybrid neural network and hyperdimensional computing (HDC)** framework for resource-efficient federated learning. Established **theoretical convergence guarantees** and designed communication-efficient strategies robust to noisy and unreliable networks. Achieved **3**× **faster convergence** and up to 2,112× lower communication cost compared to baselines ***(ACM TIoT'25)***.
- **HyperLiDAR: Efficient On-Device 3D Perception and Adaptation.** Developed a **lightweight LiDAR segmentation framework** integrating pretrained feature extraction with HDC-based online learning for efficient post-deployment adaptation. Designed a **memory-efficient buffer selection strategy** to handle large-scale point-cloud streams. Achieved **13.8**× **faster training** than SOTA methods across two benchmarks and three edge platforms ***(DAC'26)***.

Multimodal Sensing and Embedded Intelligence *Sep. 2021 – Jun. 2024*

- **Multimodal Perception and Sensor Fusion.** Developed a **vision–IMU sensor fusion** system for fine-grained gesture recognition and user interaction, supporting 12 distinct hand gestures on smartwatches ***(INFOCOM'22)***.
- **Wireless Sensing and Human–Computer Interaction.** Designed a **Wi-Fi backscatter-based lip-reading system** for touchless user authentication and semantic-level silent speech interaction, enabling novel accessibility-oriented interfaces ***(IWQoS'23)***.

· **Signal Processing and Robust Communication.** Investigated **MEMS sensor interference** and **multi-tag RF communication**, developing secure mobile sensing techniques and interference mitigation algorithms for reliable wireless communication ***(IEEE TMC'24, MobiSys'24)***.

SELECTED PUBLICATIONS

Co-first authors are marked with *. For a complete list, please refer to my [**Google Scholar**].

Multimodal AI and LLM Agents.

1. **Ye Tian***, Zihao Wang*, Onat Gungor, Xiaoran Fan, Tajana Rosing. *LifeAgentBench: Benchmarking LLMs for Long-Horizon, Cross-Dimensional Lifestyle Health Reasoning.* **EMNLP 2026**.

2. **Ye Tian**, Xiaoyuan Ren, Zihao Wang, Onat Gungor, Xiaofan Yu, Tajana Rosing. *DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs.* **IEEE MASS 2025**.

3. **Ye Tian**, Onat Gungor, Xiaofan Yu, Tajana Rosing. *Fine-grained Contextualized Activity Logs Generation based on Multi-Modal Sensor Data and LLM.* **ACM SenSys 2025**.

4. Woojin Lee, Pranav Mekkoth, **Ye Tian**, Onat Gungor, Tajana Rosing. *FoodCHA: Multi-Modal LLM Agent for Fine-Grained Food Analysis.* **IEEE ISNCC 2026**.

Autonomous Systems and Adaptive AI.

5. **Ye Tian**, Md Mizanur Rahaman, Quanling Zhao, Tianqi Zhang, Flavio Ponzina, Azad J. Naeemi, Tajana Rosing. *KROMA: Knowledge-Retaining Online Adaptation for Multimodal Fusion in Changing Driving Conditions.* **CVPR 2027 (in submission)**.

6. **Ye Tian**, Xiaoyuan Ren, Jingyi Zhang, Xiaofan Yu, Onat Gungor, Tajana Rosing. *KLDrive: Knowledge Graph-Grounded LLMs for Fine-Grained 3D Scene Reasoning in Autonomous Driving.* **AAAI 2027 (under review)**.

7. Xiaofan Yu, Yuwei Wu, Katherine Mao, **Ye Tian**, Vijay Kumar, Tajana Rosing. *DroneFL: Federated Learning for Multi-UAV Visual Target Tracking.* **IEEE T-ASE (under review)**.

Efficient and Robust Machine Learning.

8. Ivannia Gomez Moreno*, Yi Yao*, **Ye Tian***, Xiaofan Yu, Flavio Ponzina, Jingyi Zhang, Michael Sullivan, Mingyu Yang, Hun Seok Kim, and Tajana Rosing. *HyperLiDAR: Adaptive Post-Deployment LiDAR Segmentation via Hyperdimensional Computing.* **DAC 2026**.

9. **Ye Tian***, Rishikanth Chandrasekaran*, Kazim Ergun*, Xiaofan Yu, Tajana Rosing. *Federated Hyperdimensional Computing: Comprehensive Analysis and Robust Communication.* **ACM TIoT 2025**.

10. Quanling Zhao, Nilesh Prasad Pandey, **Ye Tian**, Tajana Rosing. *Superposed Inference for Hyperdimensional Computing.* **ICLR 2027 (under review)**.

11. Quanling Zhao, Jiaying Yang, **Ye Tian**, Josh Victoria, Zhijun Wang, Pietro Mercati, Onat Gungor, Tajana Rosing. *RGLD: Randomized Global-Local Density Estimation for Tabular Anomaly Detection.* **ICLR 2027 (under review)**.

Multimodal Sensing and Embedded Intelligence.

12. **Ye Tian**, Hao Zhou, Haohua Du, Chenren Xu, Jiahui Hou, Dong Ren, and Xiang-Yang Li. *BackLip: Passphrase-Independent Lip-reading User Authentication with Backscatter Signals.* **IEEE/ACM IWQoS 2023**.

13. Kaiwen Guo, Hao Zhou, **Ye Tian**, Wangqiu Zhou, Yusheng Ji, and Xiang-Yang Li. *Mudra: A Multi-Modal Smartwatch Interactive System with Hand Gesture Recognition and User Identification.* **IEEE INFOCOM 2022**.

14. Shanyue Wang, Yubo Yan, Feiyu Han, **Ye Tian**, Yuxin Ding, Panlong Yang, and Xiang-Yang Li. *MultiRider: Taming In-band Interferences in OFDM Backscatter for Parallel Communication.* **ACM MobiSys 2024**.

15. Junyang Zhang, Jiahui Hou, **Ye Tian**, Xiang-Yang Li. *WordWhisper: Exploiting Real-Time, Hardware-Dependent IoT Communication against Eavesdropping.* **IEEE Transactions on Mobile Computing (TMC) 2024**.

SELECTED INVITED TALKS

- "Efficient and Adaptive Learning for Autonomous Driving Perception," **CoCoSys Theme II Meeting**, Jul. 23, 2026.
- "Multimodal Learning with HyperLiDAR: Adaptive Post-Deployment LiDAR Segmentation via HDC," **PRISM Annual Review**, Nov. 12, 2025.
- "From Wearable Sensors to Health Reasoning with LLMs: Multimodal Activity Logging and Lifestyle QA," **CWS Research Summit**, Oct. 24, 2025.
- "Federated Hyperdimensional Computing: Comprehensive Analysis and Robust Communication," **VSAONLINE**, Oct. 20, 2025.
- "DailyLLM: Context-Aware Activity Log Generation Using Multi-Modal Sensors and LLMs," **TILOS Industry Day**, Jun. 2, 2025.

SELECTED HONORS AND AWARDS

Research Excellence: Best Poster Award, TILOS Industrial Day (Top 3, 2025); Top Ten Scientific Research Stars (Top 0.5%, 2020).

Fellowships and Scholarships: UCSD CSE Fellowship (2024); National Fellowship (Top 1%, 2023); Lixin Tang Fellowship (Top 0.2%, 2020); National Encouragement Fellowship (Top 3%, 2018, 2019, 2020); First Class Fellowship (2018, 2019, 2020).

Academic Excellence and Innovation: Outstanding Graduates of the University (Top 2%, 2021); International Mathematical Contest in Modeling, Honorable Mention (2021); Forestry Innovation and Entrepreneurship Competition, National Semi-Finalist (2020); Outstanding Representative of Innovation and Entrepreneurship (Top 0.2%, 2020); College Students Three Innovation Challenge, Provincial Second Prize (2019); Internet+ College Student Competition, Gold Award (2019).

Leadership and Service: Outstanding Student Leader in Graduate Student Union (2022); Outstanding Students & Student Leader (2018, 2019, 2020); Outstanding Representative of Social Practice (2018).

ACADEMIC SERVICE AND LEADERSHIP

Journal Reviewer: ACM Transactions on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT); ACM Transactions on Embedded Computing Systems (TECS); ACM Transactions on Internet Technology (TOIT); ACM Transactions on Sensor Networks (TOSN); IEEE Transactions on Human-Machine Systems (THMS); IEEE Transactions on Network Science and Engineering (TNSE); IEEE Internet of Things Journal (IoT-J); Engineering Applications of Artificial Intelligence (EAAI); Health Information Science and Systems (HISS); Applied Soft Computing (ASOC); Soft Computing (SC); IEEE Journal of Selected Topics in Signal Processing (JSTSP).

Conference Reviewer: ICLR; AAAI; EMNLP; ACM CHI; IEEE VR.

Session Chair: IEEE/ACM IWQoS 2023; IEEE BigCom 2022.

Research Mentorship: Research Mentor, UC San Diego ENLACE Summer Research Program (2026).

Conference Volunteer: National Forum for CS Department Chairs; Deans in Colleges and Universities.